Warm-Start Re-Optimization for Long-Fiber Raman Suppression

Fiber Raman Suppression Project

Evidence snapshot: 2026-04-28

Abstract

Long fibers are expensive to optimize because each objective and gradient evaluation requires a long propagation solve. This note documents a practical shortcut: optimize a phase mask on a cheaper short target, transfer that mask to a long-fiber grid, and then re-optimize the long target from the transferred mask instead of starting from a flat phase. In the current single-mode SMF evidence, a short-fiber mask already gives roughly −51.5 dB Raman-band fraction when evaluated at 100 m, and long-target re-optimization improves it to the mid-−50 dB range. The result is important as a strategy, but it is not yet a converged claim about the global optimum at 100 m.

Claim Status

Status: provisional but important
The warm-start/re-optimization strategy is supported by real 50–100 m runs and standard images. The long-fiber optimum is still not fully converged, so this note should be read as a research strategy note rather than a final performance benchmark.

1 Question

The practical question is simple:

> Can we make long-fiber optimization cheaper by first learning a useful phase mask on a short fiber, then using that mask as the initial guess for the long fiber?

This is different from physically re-shaping the pulse inside the fiber. The current evidence is about a *computational warm start*: the same input phase is optimized again against a longer target length. A physical segmented experiment, where a pulse is reshaped after some propagation distance, would be a different hardware problem.

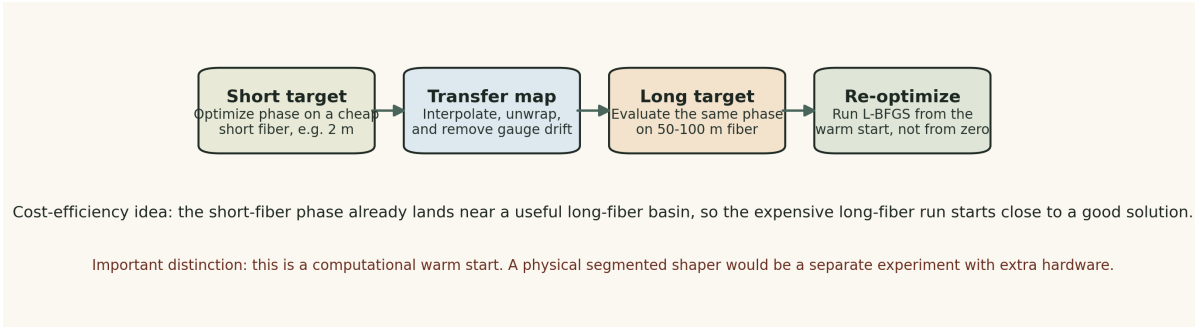

Figure 1: Warm-start re-optimization workflow. The short target gives a cheap first mask; the long target then uses that mask as an initial condition instead of beginning from zero phase.

2 Cost Function and Propagation Setup

The scalar cost is the same Raman-band energy fraction used in the rest of the single-mode optimization work:

$$J(\phi; L) = \frac{\sum_{k \in \mathcal{B}_{\mathrm{Raman}}} |\hat{u}_k(L;\phi)|^2}{\sum_k |\hat{u}_k(L;\phi)|^2}. \tag{1}$$

Most reported values use

$$J_{\mathrm{dB}}(\phi; L) = 10 \log_{10} J(\phi; L). \tag{2}$$

More negative values mean less energy in the Raman-shifted band.

The shaped input spectrum is

$$\hat{u}_k(0;\phi) = \hat{u}_{0,k} \exp(i\phi_k), \tag{3}$$

where the amplitude $\hat{u}_{0,k}$ is fixed and only the spectral phase is optimized. Long-fiber propagation then evaluates the generalized nonlinear Schrödinger model with dispersion and the delayed Raman response. The propagation details are inherited from the main single-mode model; this note is only changing the *initialization strategy* for the optimizer.

TL;DR. The optimizer is still minimizing the same Raman fraction. The only trick is that the long run starts from a phase mask that was already useful on a shorter fiber, instead of asking L-BFGS to discover the whole structure from a flat phase.

3 Transfer Map

Let ϕ_s be the phase optimized on a short fiber length L_s, and let $\omega^{(s)}$ and $\omega^{(\ell)}$ be the short and long frequency grids. The long-fiber initial phase is a transferred version of the short phase:

$$\phi^{(\ell)}_{\mathrm{init},j} = \mathcal{I}[\mathrm{unwrap}(\phi_s)]\left(\omega^{(\ell)}_j\right), \tag{4}$$

where $\mathcal{I}$ is interpolation on physical angular frequency. Frequencies outside the short-grid support are set conservatively rather than extrapolating large artificial phase structure.

After interpolation, the phase may be gauge-centered because a constant phase does not change the spectral intensity objective:

$$\phi_{\text{init}}^{(\ell)} \leftarrow \phi_{\text{init}}^{(\ell)} - \frac{1}{N}\mathbf{1}\mathbf{1}^T\phi_{\text{init}}^{(\ell)}. \tag{5}$$

The long optimization then solves

$$\phi_\ell^* \approx \arg\min_\phi J_{\text{dB}}(\phi; L_\ell), \qquad \phi^{(0)} = \phi_{\text{init}}^{(\ell)}. \tag{6}$$

Intuition Check. In linear algebra terms, this is not a new objective. It is a better starting vector. The short-fiber solution gives L-BFGS a point that already has useful destructive-interference structure, so the expensive long-fiber search spends more time refining and less time discovering the basic mask shape.

4 Experimental Method

The current long-fiber evidence uses a single-mode SMF-style setup with $L = 50$ m and $L = 100$ m targets. The 100 m configuration used a large time window and grid to avoid edge artifacts: $N_t = 32768$ and a 160 ps time window. The primary warm start came from a short-fiber optimized mask at $L_s = 2$ m.

Step	What was done
Short optimization	Obtain a phase mask on the cheap short target.
Transfer	Interpolate the unwrapped short mask onto the long-fiber frequency grid.
Warm-start evaluation	Propagate the transferred mask at the long length without updating it.
Re-optimization	Run L-BFGS on the long target, initialized from the transferred mask.
Diagnostics	Save the standard image set, energy/edge checks, final cost, gradient norm, and convergence flag.

Table 1: Minimal experimental protocol for the warm-start re-optimization lane.

5 Representative Results

Configuration	Length	Raman fraction	Status
No shaping / flat phase	100 m	−0.20 dB	control
Transferred short-fiber mask, no re-optimization	100 m	−51.50 dB	warm-start only
Long-fiber re-optimization from warm start	100 m	−54.77 dB	archived run, not converged
Long-fiber re-optimization from warm start	100 m	−55.92 dB	later rerun, not converged
No shaping / flat phase	50 m	−1.91 dB	control
Transferred short-fiber mask, no re-optimization	50 m	−51.92 dB	warm-start only
Long-fiber re-optimization from warm start	50 m	−60.74 dB	short re-opt run

Table 2: Current evidence for long-fiber warm starts. The main supported claim is not that the long-fiber optimum is finished; it is that the short-fiber mask lands extremely close to a useful long-fiber basin.

At 100 m, the warm start alone gives about 51 dB improvement over the flat phase control. Re-optimization gives another 3–4 dB in the available runs. At 50 m, re-optimization gave a larger extra improvement, about 8.8 dB beyond the transferred mask. This supports the view that warm starts are valuable, but the amount of extra gain depends strongly on length and on whether the long optimization has actually converged.

Provenance Check

The numbers and images in this note are based on saved long-fiber artifacts and standard image sets. The strongest table entries are the 100 m no-shaping control, the transferred-mask evaluation, and the warm-started re-optimization reruns. The important caveat is convergence: the available 100 m re-optimization runs are useful evidence for the strategy, but their gradient norms are not small enough to call the final masks fully converged optima.

TL;DR. This note is trustworthy as a compute-strategy result: warm starts clearly land near a useful long-fiber basin. It is not yet a final benchmark claiming the best possible 100 m waveform.

6 No-Optimization Control

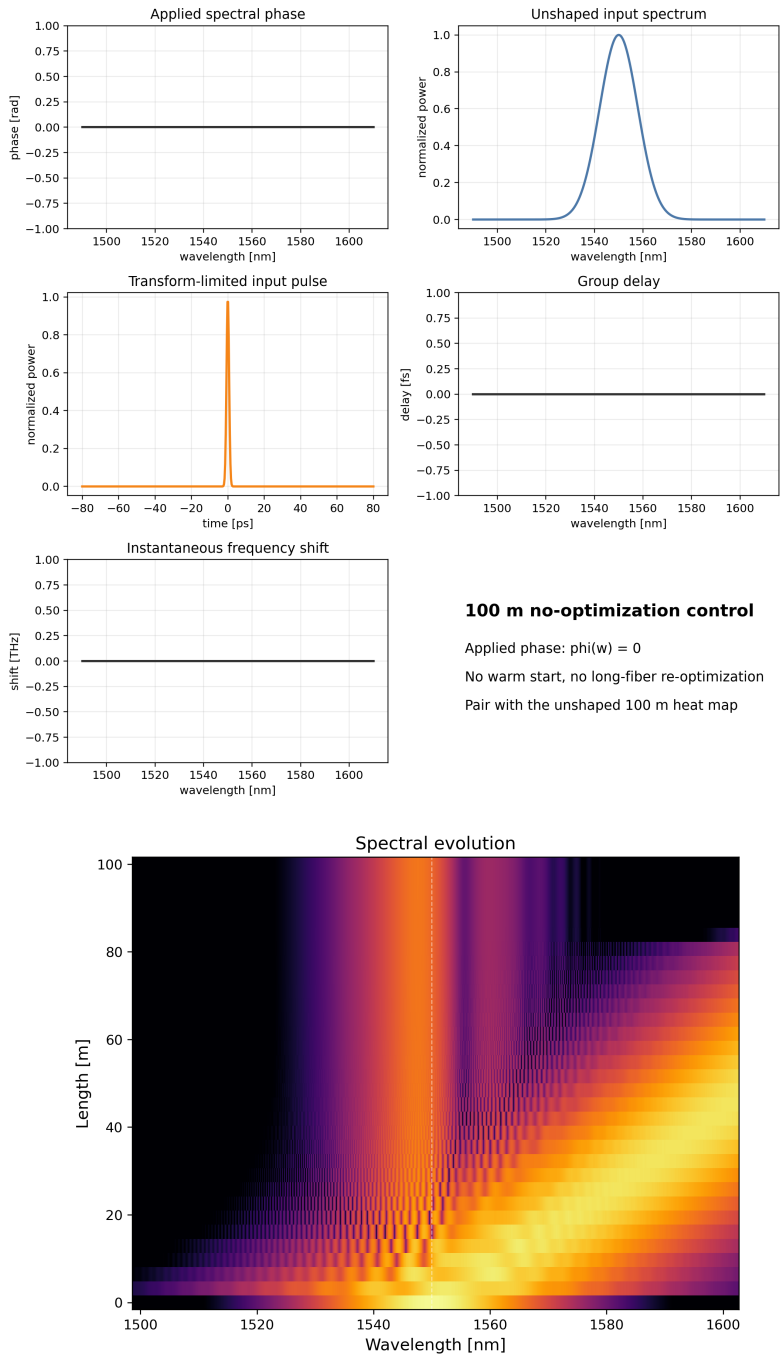

Figure 2: No-shaping control for the 100 m long-fiber setting. The phase diagnostic shows that no spectral phase was applied; the heat map is the reference case for judging whether the optimized and warm-started masks are doing real Raman suppression rather than only changing the plotted scale.

7 Optimized Long-Fiber Example

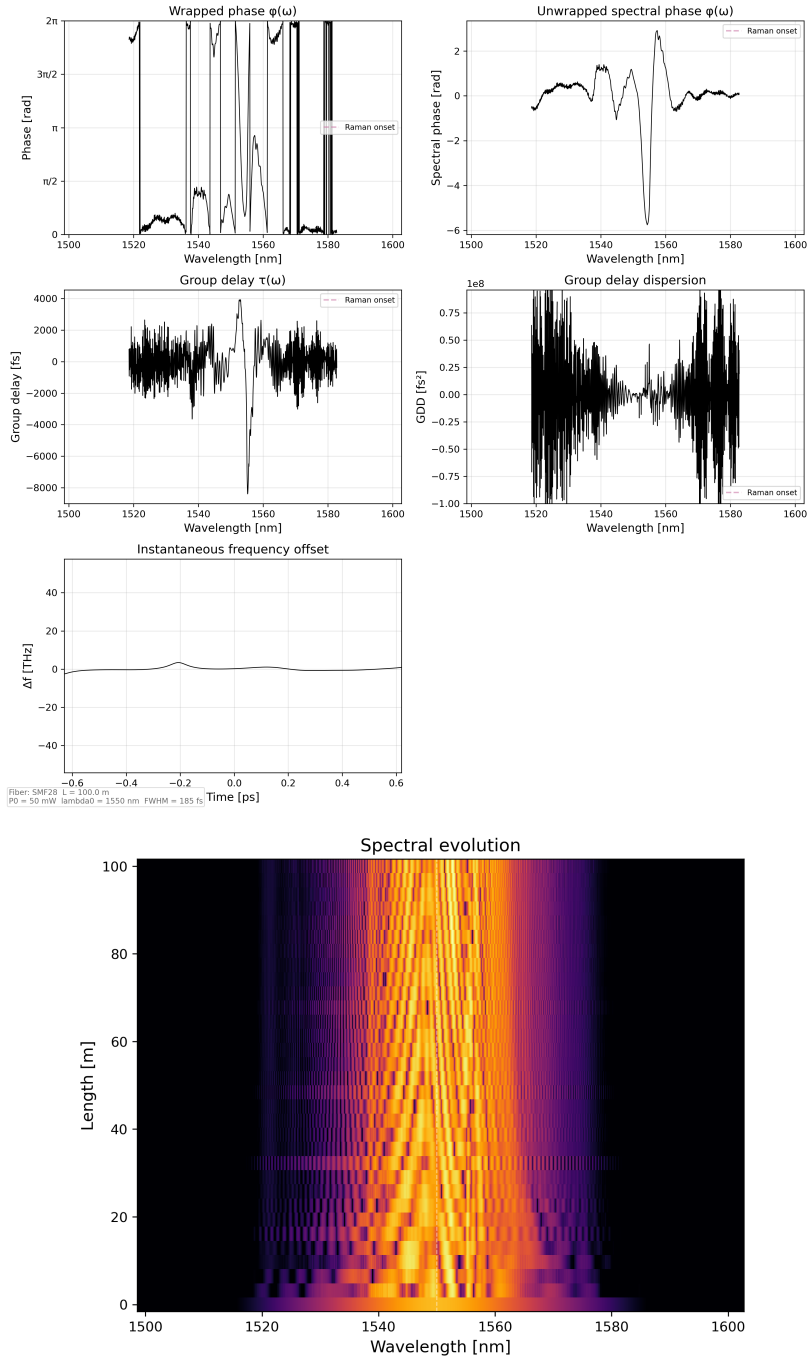

Figure 3: Warm-start re-optimized 100 m example. The diagnostic is useful, but the profile is visibly rough; this is why the result is presented as a compute-strategy result rather than a lab-ready waveform prescription. The matching heat map is the main visual evidence that the phase mask changes the Raman transfer path.

8 Interpretation

The result suggests a useful split between *basin finding* and *basin refinement*. The short-fiber mask appears to contain much of the structure needed to avoid Raman transfer even over much longer propagation. The long-fiber re-optimization then adjusts that structure for the specific long target.

The long-fiber phase is not safely described as "just a quadratic chirp." The recorded quadratic-fit checks found poor polynomial fit quality over the amplitude-weighted pulse band, and pure dispersion-length scaling did not predict the long-fiber phase coefficient. That matters because it means the warm start is carrying nonlinear structural information, not only a simple GVD pre-compensation.

TL;DR. The short mask is not magic, and it is not guaranteed to be the final answer. It is a strong first guess. For expensive long fibers, that may be the difference between a tractable optimization and a blind search.

9 Presentation Capsule

If this note becomes a long-fiber strategy slide, the clean message is:

- Long-fiber optimization is expensive, so initialization matters.
- A short-fiber optimized mask can act as a warm start for the long target, then the optimizer adapts it to the longer propagation length.
- The visual story should be the 100 m no-shaping control versus the optimized long-fiber heat map, not an internal run-history label.
- The result is a practical strategy note, not proof of a globally optimal 100 m waveform.

10 Limitations

- The 100 m re-optimized runs did not satisfy a strict convergence claim; the final gradient norms remain too large to call this a completed optimum.
- The evidence is single-mode. Multimode long-fiber optimization should not inherit these results automatically.
- The current note does not prove that warm-starting is cheaper than every possible cold-start strategy; it shows that the warm start reaches a very good basin and gives a practical compute path.
- A real segmented hardware experiment would need in-line shaping or some other way to modify the field after partial propagation. The current workflow only changes the numerical optimizer initialization.
- The 100 m phase diagnostic is rough. That is acceptable for a research strategy note, but not for a final lab waveform prescription.

11 Reproduction Capsule

For a clean reproduction, run on the heavy-compute machine with Julia threading enabled. The relevant public workflow surface is in `scripts/research/longfiber/`:

- `longfiber_setup.jl`: constructs the long-fiber problem while honoring the requested grid and time window.
- `longfiber_optimize_100m.jl`: runs the warm-started long-fiber L-BFGS optimization.
- `longfiber_validate_100m.jl`: checks the final result and reports the evidence numbers.
- `longfiber_regenerate_standard_images.jl`: regenerates the standard diagnostic image set when needed.

The required output artifacts are the final cost, convergence flag, gradient norm, energy drift, boundary/edge fraction, and the standard image set: phase profile, phase diagnostic, optimized evolution, and unshaped evolution. The note should not be upgraded from provisional to production-ready until a fresh rerun confirms these artifacts and the PDF figures are visually checked again.

12 Sources

- G. P. Agrawal, *Nonlinear Fiber Optics*, 6th ed., Academic Press, 2019. Standard reference for the generalized nonlinear Schrödinger model, Kerr nonlinearity, dispersion, and stimulated Raman scattering.
- J. Hult, "A fourth-order Runge–Kutta in the interaction picture method for simulating supercontinuum generation in optical fibers," *Journal of Lightwave Technology*, 25(12), 3770–3775, 2007. https://doi.org/10.1109/JLT.2007.909373
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